

Kanokpon Mettasat

Electrical & Computer Engineering, Honors Program · Virginia Military Institute, Class of 2027

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SUMMARY

Electrical & Computer Engineering honors student (GPA 3.79 / 4.00 major) who ships production software end-to-end — FastAPI services, a multi-tenant LINE bot, and a self-hosted analytics dashboard — and runs independent quantitative-trading research with pre-registered hypotheses, walk-forward validation, and explicit bias auditing. Seeking a software / test / research engineering role where rigor and ownership matter.

EDUCATION

Virginia Military Institute

Aug 2023 – May 2027 (expected)

B.S., Electrical and Computer Engineering — Institute Honors Program

- **GPA 3.79 overall / 4.00 in major**; Dean's List, multiple terms; Honors scholarship since 2024.
- Tau Beta Pi — Engineering Honor Society, Virginia Chapter (inducted Nov 2025); conference representative (Mar 2026).
- IEEE SoutheastCon 2026 — **top 7 of 40 teams, Undergraduate Circuit Design** (Mar 2026).
- Royal Thai Army Cadet School, Class 74 — **selected #1 of 348** for the army's study-in-US program (2022).

TECHNICAL SKILLS

Languages: Python, JavaScript (ES2020), TypeScript, PowerShell, Bash

Web / API: FastAPI, Jinja2, HTML5 / CSS3, REST

Data / Quant: pandas, NumPy, MetaTrader 5 API, Probabilistic Sharpe / Deflated Sharpe

Tooling: pytest, pyright, Playwright, Git, GitHub Actions

Infrastructure: Windows Server, Tailscale Funnel, Scheduled Tasks, SQLite, LINE Messaging API

SELECTED EXPERIENCE

Self-Hosted Analytics & Trading Dashboard

Python · FastAPI · Jinja2 · Playwright

- Built and operate a 12-page FastAPI/Jinja2 dashboard on a Windows VPS with a **~679-test** suite, leak-guard tripwires, and zero-downtime redeloys.
- Shipped this public portfolio as an isolated service proven free of any private data by an automated leak-test gate.

Quantitative Trading Research Framework

Python · pandas · MetaTrader 5 · pytest

- **60+ backtests** across FX, metals, indices, and crypto with pre-registered hypotheses, walk-forward out-of-sample validation, and probabilistic / deflated Sharpe gates.
- Companion PRISMA-style literature review over **63 IEEE papers** (manuscript in preparation).

Multi-Tenant LINE Official-Account Bot

Python · FastAPI · SQLite · LINE API

- Multi-owner messaging bot with scheduled reminders, daily SQLite snapshots, and a self-healing restart loop, running unattended (**~201 tests**).

LEADERSHIP

- OGA Representative, Hotel Company — leads new-cadet orientation (2026); Old Corps Corporal & MathLab tutor.
- Only Thai cadet at VMI; Expert rifle classification; commendation from the Chief of Defence Forces (2024).

Selected Projects

Six of the systems behind the résumé — full write-ups and live demos at km-portfolio.onrender.com/portfolio



Trading-Research Dashboard

A 12-page FastAPI/Jinja2 monitoring and decision-support dashboard on a Windows VPS: project tracking, an isometric office view of research agents, a live-demo strategy monitor, and a Smart Money Radar scoring 13F / Form-4 signals.

Python FastAPI Jinja2 Phaser 3 pytest

~679 tests · 12+ pages · zero-downtime redeploy



Quantitative Trading Research

60+ backtests across FX, metals, indices and crypto with pre-registered hypotheses, walk-forward out-of-sample validation, bias auditing, and probabilistic / deflated Sharpe gates.

Python pandas MetaTrader 5 pytest

60+ backtests · PRISMA review of 63 IEEE papers



Live Web Karaoke — Group Mode

A no-install PWA that turns a phone into a karaoke mic and streams the vocal to a TV over WebRTC with a QR-join flow. Core engineering problem: driving the receiver jitter buffer down to kill echo.

JavaScript WebRTC PeerJS PWA

Live on Vercel · v0.5.1 · karaoke-web-prototype.vercel.app



Money Compass — Wealth Aggregator

A read-only personal-finance aggregator pulling bank and card transactions via SimpleFIN into cashflow, runway and subscription views. Auth-gated, fails closed, and never renders figures on a public surface.

Python FastAPI SQLite SimpleFIN uPlot

40+ tests · auth-gated + leak-guarded



SURI — Systematic Literature Review

A pre-registered, PRISMA-style systematic mapping of quantum-computing threats to blockchain across three research questions, with a multi-dimensional taxonomy and coverage matrix.

IEEE Xplore PRISMA Taxonomy Python

63 IEEE papers · 3 RQs · manuscript in prep



Edge Factory — Research Automation

A Python orchestrator that runs each strategy candidate down a nine-station conveyor — seed, expand, dedup, prescreen, backtest, bias-audit, gate, dossier, verdict — so ideas are killed cheaply and early.

Python pandas pytest pyright

MVP v0.1 · 36 tests · pyright 0